

RiskPulse

An AI/NLP risk engine that turns news and tweets into structured financial risk signals

Avijit Dutta

Vellore Institute of Technology, Vellore

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 **56,931**

signals produced

 **89%**

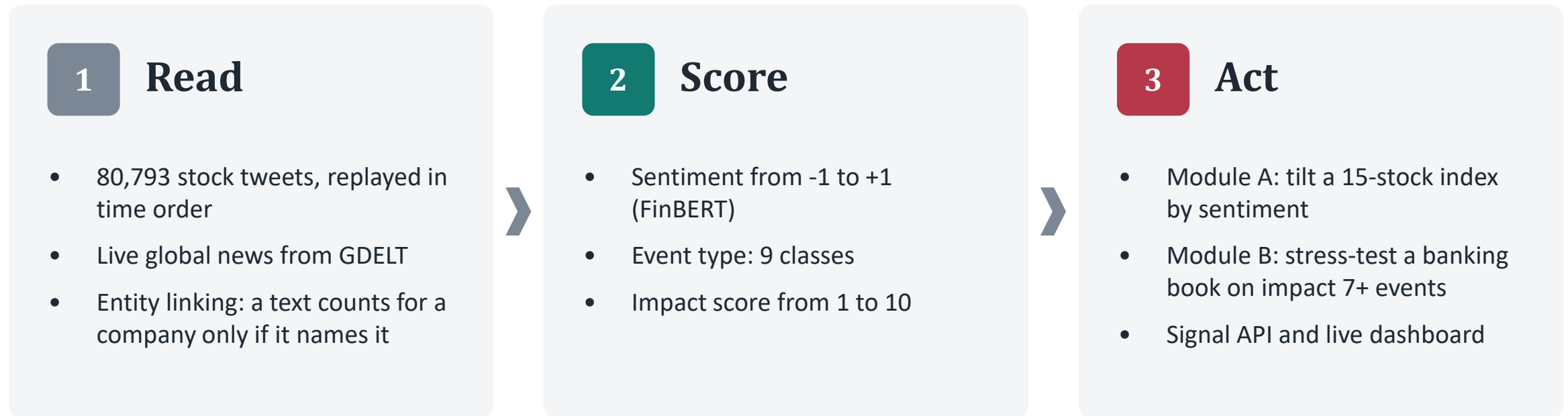
event-type accuracy

 **2.9x**

normal price move on high-impact days

Risk appears in text before it appears in prices

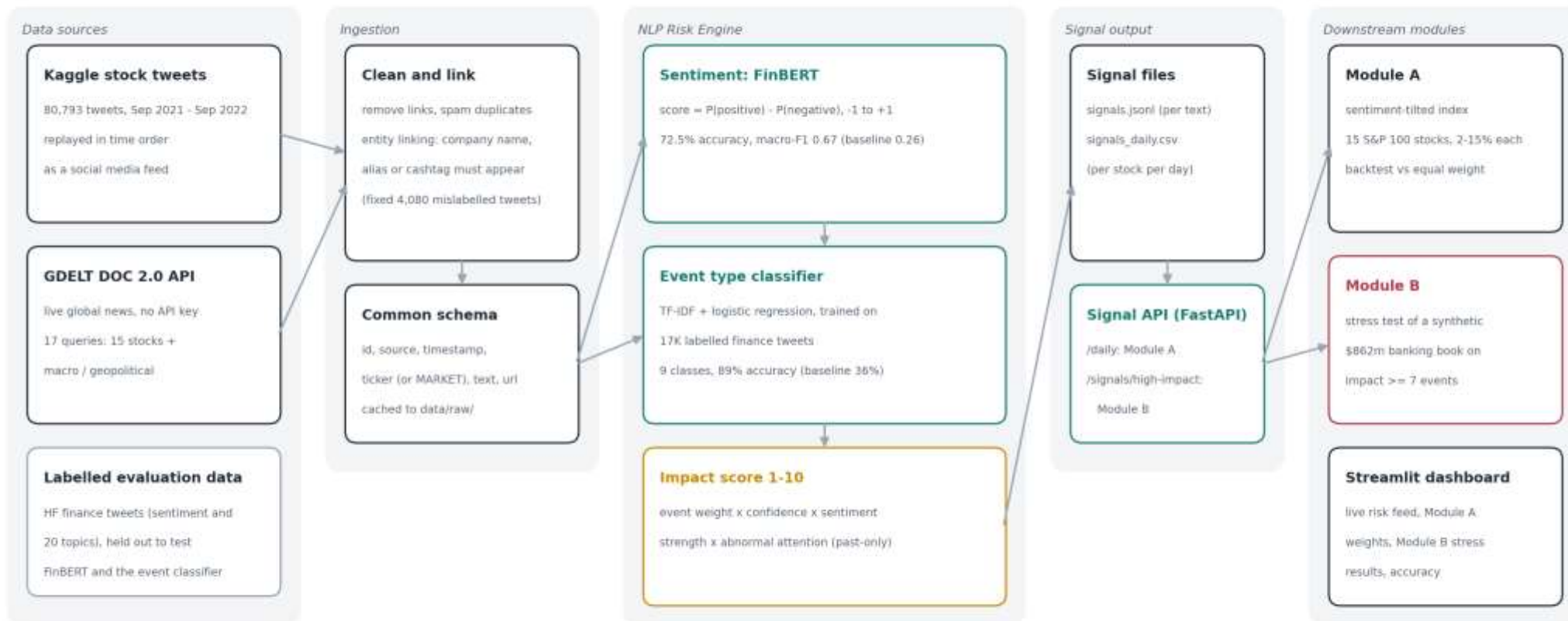
A downgrade, a lawsuit, a Fed comment or a sanctions headline is written down before the market fully reacts. Risk teams cannot read thousands of posts an hour, so the early signal is missed.



System design: one engine, two consumers

RiskPulse: AI/NLP Risk Engine architecture

Unstructured text in, structured risk signals out, consumed by two downstream modules



Validation against real prices

Stock-days the engine scores 7-10 moved 2.9x their normal amount the same day (0.96x for scores 1-3).

Module A found no next-day edge before costs: the engine is a risk monitor, not a trading signal. Module B: a +2% rate shock is the banking book's biggest risk (-10%).

Implementation choices, and why

Entity linking

A name, alias or cashtag must appear. It caught a data bug: 4,080 tweets copied under three companies.

FinBERT sentiment

Pre-trained on financial language. Tested on tweets it never saw, not on PhraseBank, which it was trained on.

Event classifier

TF-IDF + logistic regression: trains in a minute and labels 57K texts in seconds on a CPU. Zero-shot would take hours.

Impact score

Event severity x confidence x sentiment strength x abnormal attention, using past days only (no look-ahead).

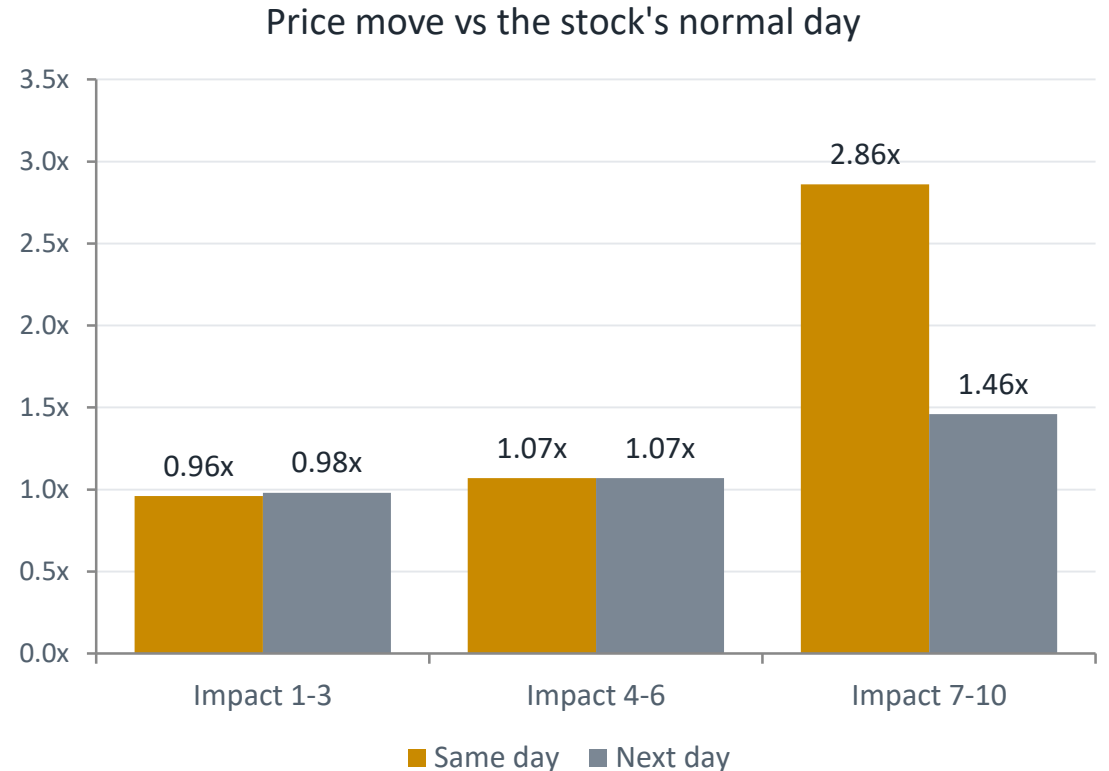
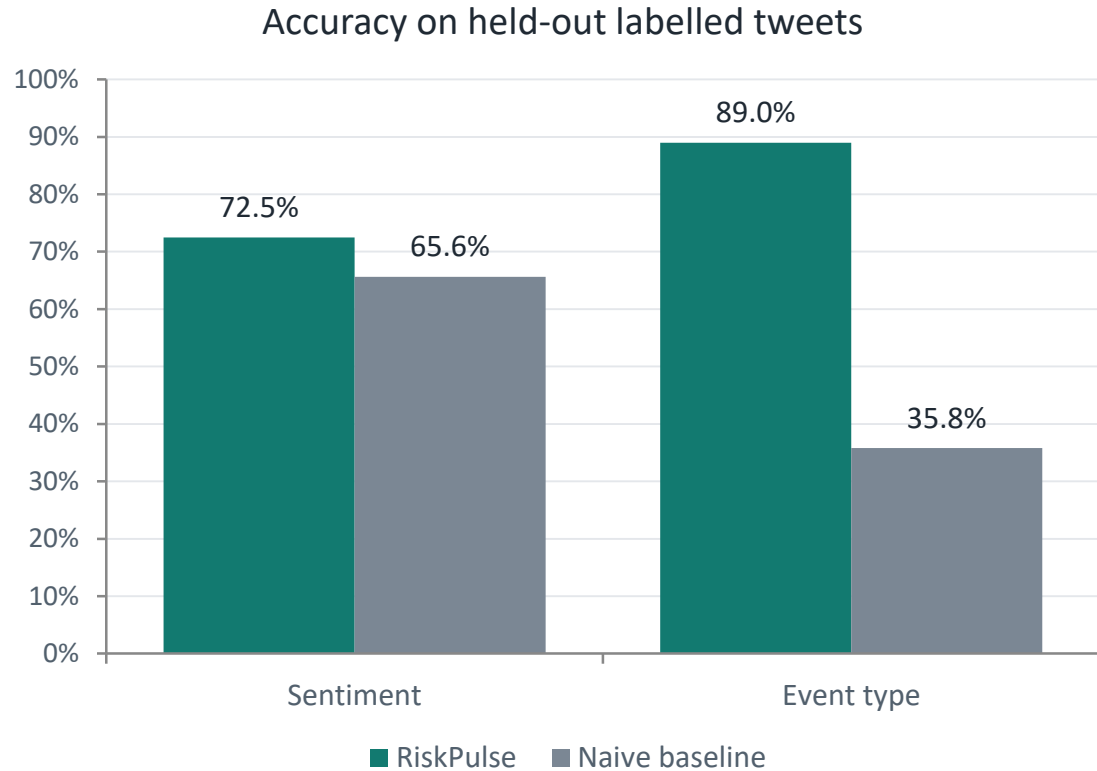
Signal API

FastAPI endpoints: Module A subscribes to /daily, Module B to /signals/high-impact.

Honest backtests

Equal-weight benchmark, 10 bps trading costs, and a 9-setting robustness grid.

Every component beats its naive baseline on unseen data



56,931 signals from 56,819 cleaned tweets and live GDELT headlines. Macro-F1: sentiment 0.67 (baseline 0.26), events 0.87. The impact score ranks price moves better than tweet volume alone (0.085 vs 0.059).

A real-time risk monitor, not a trading signal

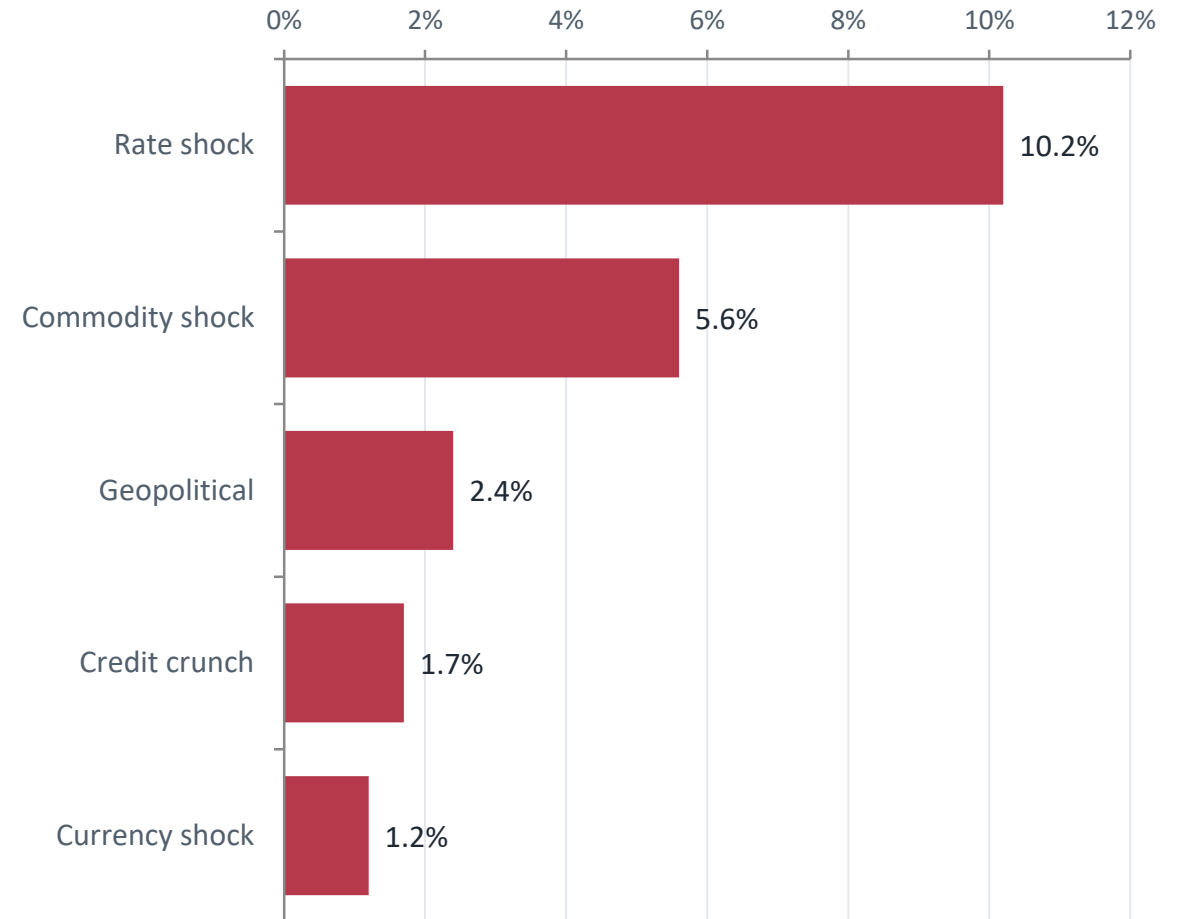
A Module A: sentiment-tilted index

Matched equal weight before costs (+0.03% a year) in the 2022 bear market; 10 bps costs left it 0.6% a year behind. Tweets mostly react to moves that already happened.

B Module B: event-driven stress tests

40 stress tests triggered on a synthetic \$862m banking book. Example: Chair Powell signals a 50 bp hike (21 Apr 2022) and the rate-shock scenario costs \$43.8m. Every result traces back to its headline.

Loss on the \$862m book (impact-8 event)



Limitations and next steps

Limitations

- FinBERT reads analyst jargon ("cuts to Equal Weight") as neutral
- Word-based event model is fooled by slang ("Gold secured! \$TSLA")
- Event weights are hand-set; only 49 stock-days scored 7+
- Stress shocks are fixed per scenario, first-order valuation

Next steps

- Fine-tune FinBERT on labelled finance tweets
- Learn impact weights from realised price moves
- Size each shock to the specific event; full repricing
- Streaming news feed instead of rate-limited GDELT polling

Code, data and demo: github.com/<avijitdutta1230000-crypto>/VIT_Vellore-Avijit_Dutta-hackathon